

CIVIC SENTINEL

Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd.

1. Abstract

The Members of Parliament Local Area Development Scheme (MPLADS) handles numerous developmental projects implemented across various administrative authorities and agencies. Due to the high volume and complexity of the financial, execution, and project data, identifying anomalies, cost deviations, delays, or duplicated works early is a significant challenge.

MPLADS Sentinel is an AI-powered early-warning and risk intelligence system built to operate alongside the existing e-SAKSHI ecosystem. Instead of acting as a replacement, it continuously analyzes financial and project data to detect risks, generating aggregated project-level risk scores. The platform emphasizes explainable risk intelligence by presenting the contributing factors and evidence alongside a recommended verification action. The system uses a human-in-the-loop methodology where AI prioritizes and flags potential risks, but authorized human officials perform the actual investigation and verification.

2. Introduction & Problem Statement

- **Problem Statement ID:** SIH26102
- **Organization:** Ministry of Statistics and Programme Implementation (MoSPI).
- **Department:** Data Informatics & Innovation Division (DIID).
- **Category:** Software.
- **Theme:** Smart Automation / Miscellaneous.

The core problem defined by SIH26102 is the need for an AI-powered system capable of detecting inefficiencies, anomalies, and fraud in the implementation of the MPLAD Scheme. Current project monitoring requires authorities to manually examine every project record, which makes it difficult to spot unusual financial behaviors, prolonged inactivity, or payment-progress mismatches. The proposed system is required to generate decision-support dashboards, risk-based alerts, predictive insights, and early-warning mechanisms for authorities including Members of Parliament, District Authorities, State Nodal Authorities, and the Ministry.

3. Existing System vs. Proposed Solution (MPLADS Sentinel)

The existing MPLADS digital infrastructure (e-SAKSHI) is primarily operational, providing capabilities for work sanctioning, payments, and workflow tracking. MPLADS Sentinel acts as an intelligence layer above this infrastructure.

- **Existing Ecosystem:** Records and tracks information, displays operational data, and heavily relies on manual examination to find unusual patterns.

- **MPLADS Sentinel:** Analyzes the recorded information, predicts emerging problems, highlights unusual patterns, and prioritizes where human attention is required.

The conceptual shift is moving from reactive monitoring (waiting for a project to fail) to proactive, risk-based monitoring (identifying risks early and alerting authorities to intervene). Furthermore, the system includes "Estimate Intelligence," which calculates a baseline estimate for a project before approval using project parameters and rate data, comparing it against the submitted estimate to prevent excessive funding commitments.

4. System Architecture and Workflow

The system architecture operates as a continuous risk-intelligence loop.

1. **Data Source:** Ingests information from e-SAKSHI/MPLADS (mocked via representative data in the prototype).
2. **Data Processing:** Handles cleaning, validation, and feature engineering.
3. **AI/ML + Rule Engine:** Analyzes risks across multiple dimensions including cost, delay, duplicates, and activity.
4. **Risk Scoring:** Generates a dynamic 0–100 overall risk score for the project.
5. **Explainable AI:** Translates the score into understandable reasons, evidence, and recommendations.
6. **Alert & Assignment:** Routes role-based alerts to the correct authority.
7. **Decision Dashboard:** Displays risk, timeline, financial maps, and project intelligence.
8. **Human Review & Action:** Authorized personnel verify, investigate, and resolve the alerts.
9. **Feedback/Outcome Loop:** Verified outcomes flow back into the system to improve future models and rules.

5. AI/ML Risk Detection Methodology

The risk detection engine does not rely on a single parameter but aggregates multiple signals to identify projects requiring attention.

- **Delay Risk Detection:** Compares expected physical progress against actual progress based on the elapsed project timeline to predict delay probabilities.
- **Cost Risk & Financial Anomalies:** Evaluates financial trajectories to find cost overruns or unusual expenditure behaviors. It utilizes unsupervised anomaly detection models like Isolation Forest to identify statistically unusual project patterns.
- **Duplicate/Similarity Risk:** Utilizes Natural Language Processing (NLP) techniques, such as TF-IDF or Sentence Transformers, alongside location and category data, to identify highly similar or potentially overlapping projects.
- **Activity/Inactivity Risk:** Flags projects that lack meaningful progress updates for prolonged periods.
- **Payment–Progress Mismatch:** Highlights cases where financial payouts are significantly higher than the reported physical progress.
- **Geospatial Intelligence (GIS):** Evaluates project locations using coordinates to find spatial clustering, nearby similar works, and to provide terrain logistics context.

Important Principle: A risk signal or anomaly detected by AI does not equate to confirmed fraud. The platform highlights potential issues for human verification rather than making autonomous accusations.

6. Technology Stack

The platform uses a modular, scalable technology stack:

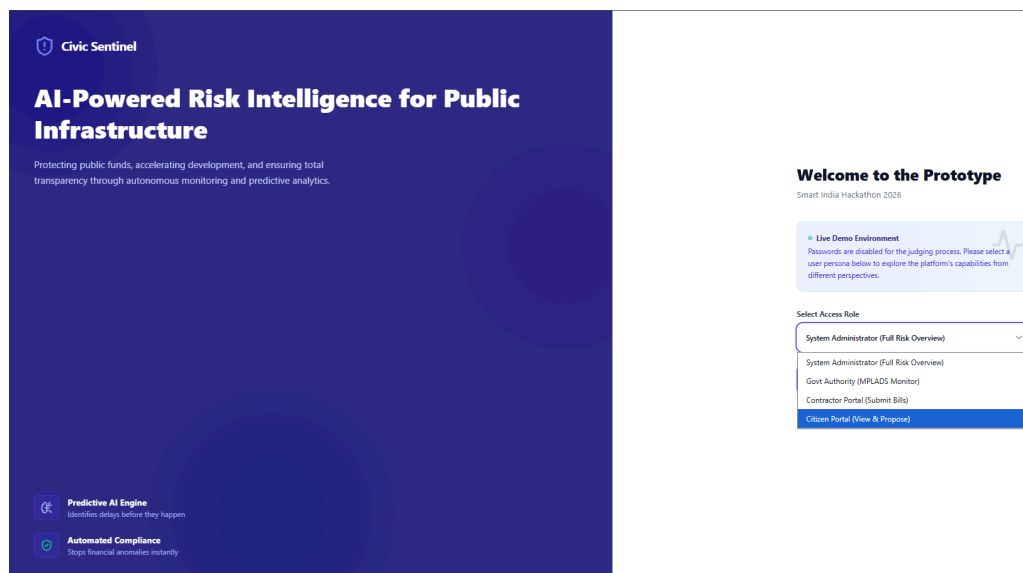
- **Frontend:** React.js, Tailwind CSS for the dashboard.
- **Maps/GIS:** Leaflet, PostGIS.
- **Backend & APIs:** Python, FastAPI, REST APIs.
- **Database:** PostgreSQL, pgvector (for vector embedding storage).
- **AI/Machine Learning:** Python, Scikit-learn, XGBoost, Sentence Transformers.
- **Document Intelligence (RAG):** LlamaIndex, PyMuPDF, OCR.
- **Deployment:** Docker, AWS/GCP.

7. Prototype Demonstration & Interface Screenshots

(Note: As requested, the following section highlights the visual workflows of the civic-sentinel-ai.vercel.app demo environment, documenting the critical user interfaces necessary to manage MPLADS risk).

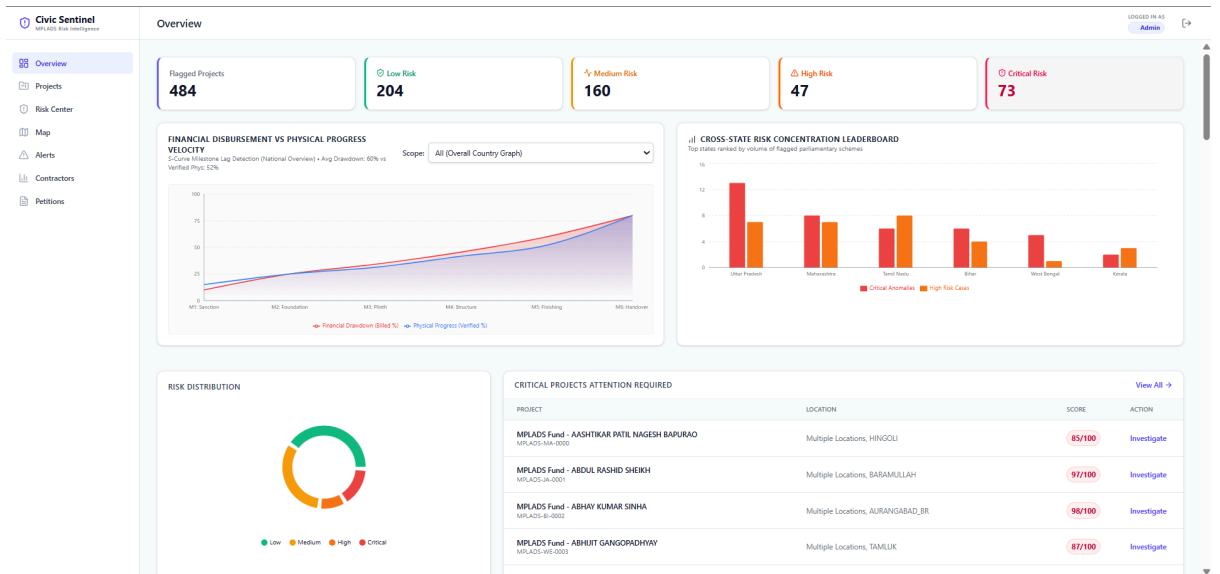
7.1 Role-Based Access and Dashboard Overview

The platform supports role-based access to ensure users only see information relevant to their clearance (e.g., Ministry Admin, District Authority, MP, Implementing Agency, and Public).



Description: The login interface of the demo website (civic-sentinel-ai.vercel.app) displaying the drop-down menu for user personas: System Administrator, Govt Authority, Contractor Portal, and Citizen Portal.

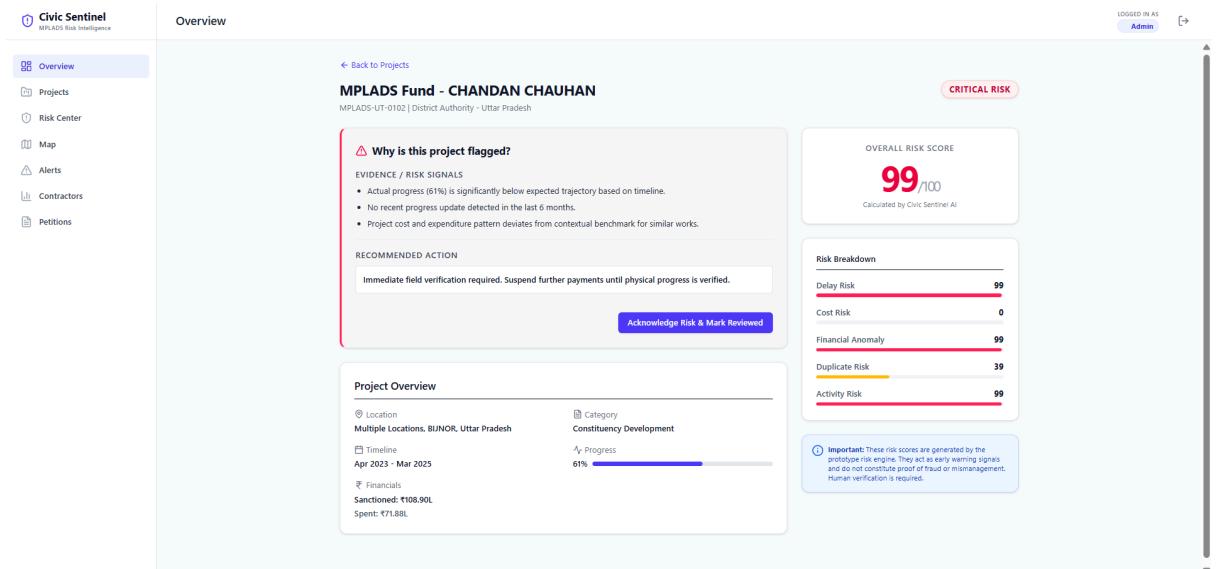
The main overview dashboard provides a high-level view of the monitored project portfolio.



Description: The main dashboard interface showing key performance indicators (KPIs) such as Total Flagged Projects, Low/Medium/High/Critical Risk counters, and a Risk Distribution pie chart. A sidebar menu provides navigation to Projects, Risk Center, Map, and Alerts.

7.2 Project Intelligence & Explainable AI

When a user drills down into a specific project, the system provides a multi-dimensional risk breakdown. Crucially, the system explains *why* a score was given.



Description: A detailed view of a single MPLADS project (e.g., MPLADS Fund - CHANDAN CHAUHAN). The screen displays an "Overall Risk Score" of 99/100.

Why is this project flagged?

EVIDENCE / RISK SIGNALS

- Actual progress (61%) is significantly below expected trajectory based on timeline.
- No recent progress update detected in the last 6 months.
- Project cost and expenditure pattern deviates from contextual benchmark for similar works.

RECOMMENDED ACTION

Immediate field verification required. Suspend further payments until physical progress is verified.

Acknowledge Risk & Mark Reviewed

Description: The "Why is this project flagged?" panel. It lists Evidence/Risk Signals (e.g., "Actual progress is significantly below expected trajectory", "No recent progress update detected in the last 6 months"). It also provides a Recommended Action (e.g., "Immediate field verification required").

7.3 Risk Center & Prioritization

The Risk Center allows officials to stop asking "Which projects exist?" and start asking "Which projects require attention first?"

Overview

LOGGED IN AS Admin

Critical Risk Projects73

High Risk Projects47

Elevated Risk (Monitor)160

Active Risk Directory

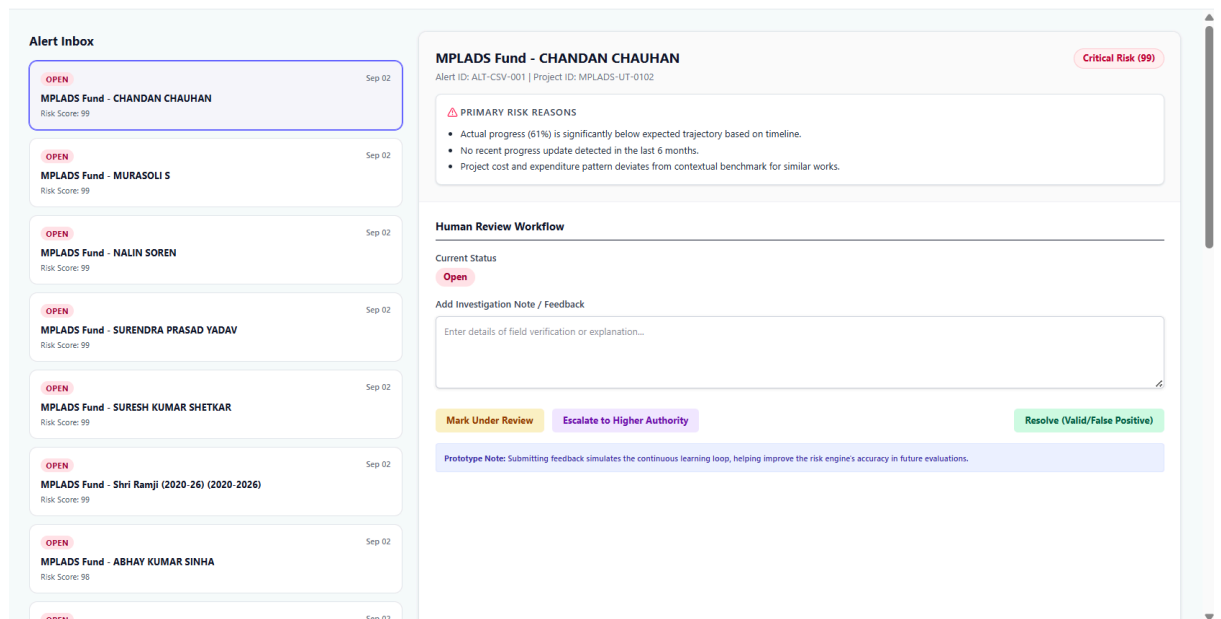
Search risky projects...

Project Name	Location	Risk Score	Primary Risk Factor	Status	Action
MPLADS Fund - CHANDAN CHAUHAN	BIJNOR, Uttar Pradesh	99/100	DELAY	Active Alert	Investigate
MPLADS Fund - MURASOLI S	THANJAVUR, Tamil Nadu	99/100	DELAY	Active Alert	Investigate
MPLADS Fund - NALIN SOREN	DUMKA(ST), Jharkhand	99/100	DELAY	Active Alert	Investigate
MPLADS Fund - SURENDRA PRASAD YADAV	JAHANABAD, Bihar	99/100	COST	Active Alert	Investigate
MPLADS Fund - SURESH KUMAR SHETKAR	ZAHIRABAD, Telangana	99/100	DELAY	Active Alert	Investigate
MPLADS Fund - Shri Ramji (2020-26) (2024-2030)	Uttar Pradesh (Rajya Sabha - Elected MP), Uttar Pradesh	99/100	DELAY	Active Alert	Investigate
MPLADS Fund - ABHAY KUMAR SINHA	AURANGABAD_BR, Bihar	98/100	DELAY	Active Alert	Investigate
MPLADS Fund - Jai Prakash	HARDOI (SC), Uttar Pradesh	98/100	DELAY	Active Alert	Investigate
MPLADS Fund - SALENG A SANGMA	TURA, Meghalaya	98/100	DELAY	Active Alert	Investigate
MPLADS Fund - SANJAY DINA PATIL	MUMBAI NORTH EAST, Maharashtra	98/100	DELAY	Active Alert	Investigate
MPLADS Fund - Shri Banshilal Gurjar (2024-30) (2024-2030)	Madhya Pradesh (Rajya Sabha - Elected MP), Madhya Pradesh	98/100	DELAY	Active Alert	Investigate

Description: A prioritized directory of risky projects ranked by their risk scores. It lists projects, their locations, the primary risk factor (e.g., DELAY, COST), and provides quick action buttons to "Investigate".

7.4 Alerts & Human Verification Workflow

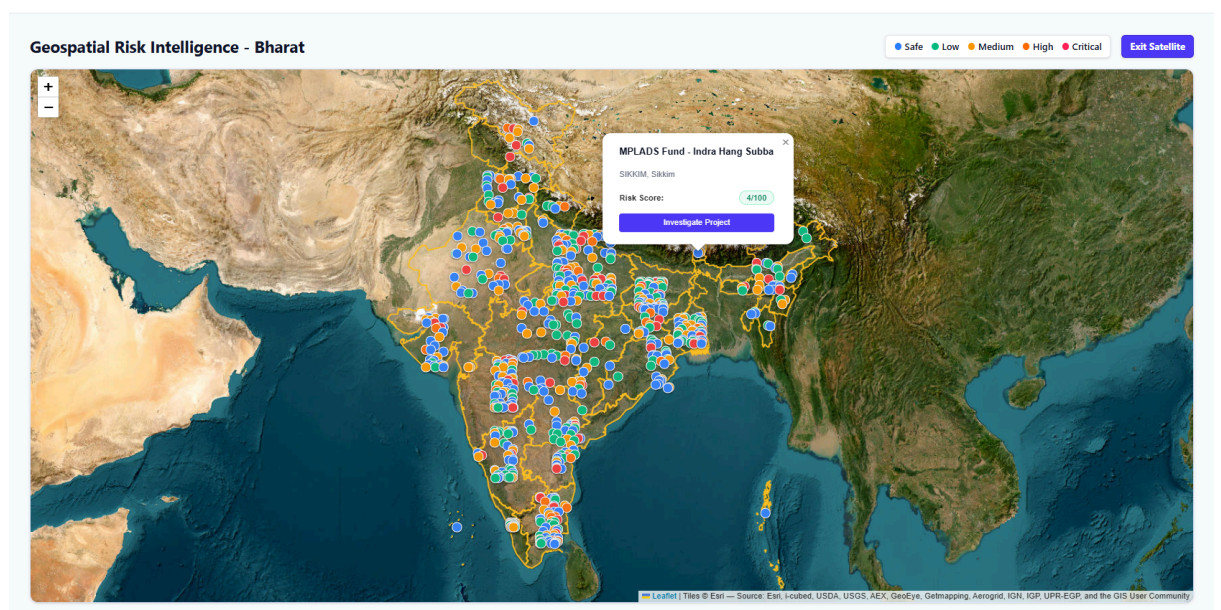
Alerts are routed to the responsible administrative authority. The system uses a Human-in-the-Loop workflow where the official reviews the AI's findings and inputs their decision.



Description: An interface showing the Alert Inbox on the left. The main panel shows a critical alert and features a "Human Review Workflow" section with a text box for adding investigation notes, alongside buttons to "Mark Under Review", "Escalate to Higher Authority", or "Resolve (Valid/False Positive)".

7.5 GIS-Based Risk Monitoring

Geospatial mapping allows authorities to view projects and their associated risk levels geographically, helping to identify clusters of similar works or high-risk zones.

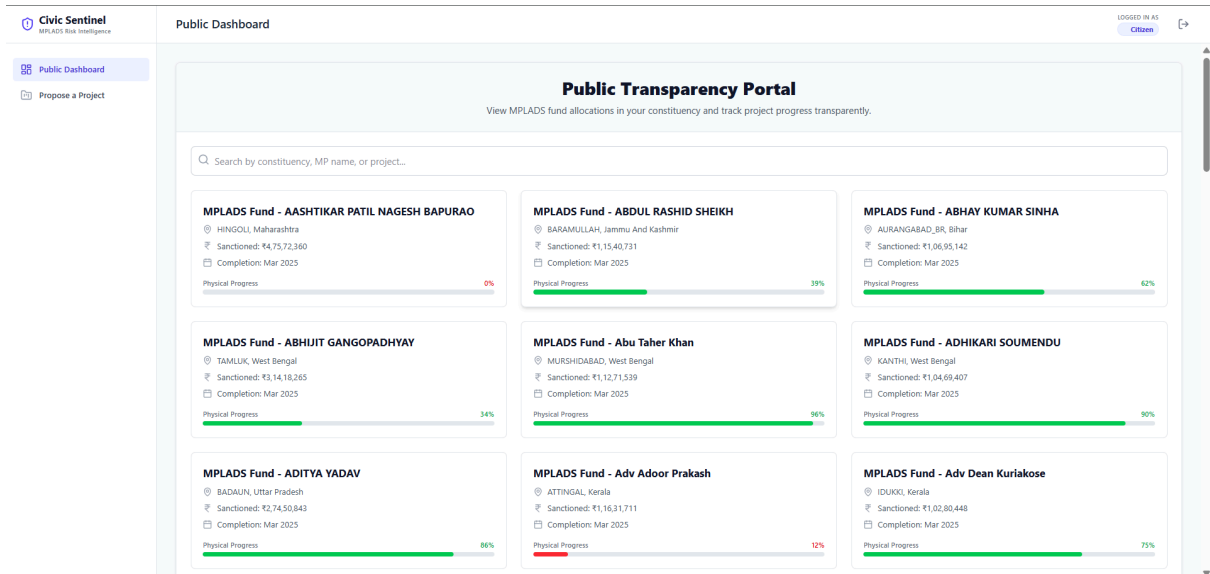


Description: The Map module interface displaying a geographical view of India

with project markers. The markers are color-coded based on risk levels to support location-based monitoring and inspection planning.

7.6 Citizen Transparency Portal

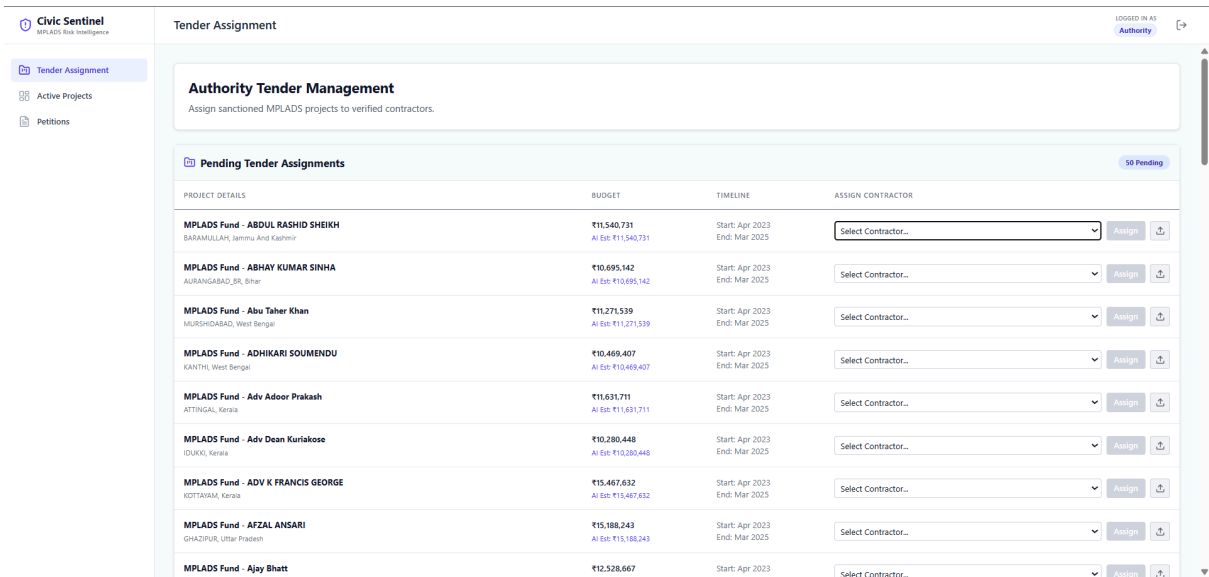
A public-facing view that omits sensitive internal investigation notes but shares general project progress, location, and fund allocation to promote transparency.



Description: The Citizen view showing standard project cards with basic details: Project Name, Location, Sanctioned Amount, Expected Completion Date, and a Physical Progress bar.

7.7 Govt Authority (MPLADS Monitor) Portal

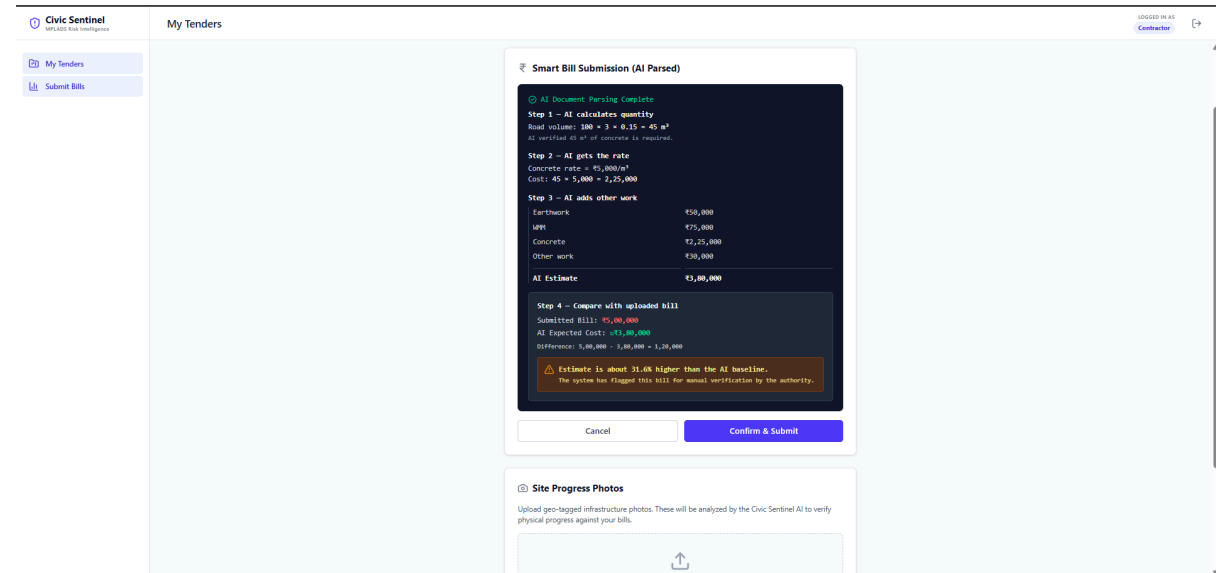
The Govt Authority view provides operational control for district-level officials, allowing them to manage tender assignments and verify local projects.



Description: The Govt Authority interface showing specific administrative capabilities, such as Pending Tender Assignments, allowing officials to assign sanctioned MPLADS projects to verified contractors.

7.8 Contractor Portal (Submit Bills)

The Implementing Agency/Contractor view allows executing bodies to monitor the projects under their responsibility, track funding, and submit progress.



Description: The Contractor Portal interface displaying a summary of Active Tenders, alongside details for assigned projects including Sanctioned budget, Billed amounts, and Pending Funds.

8. Feasibility, Impact, and Benefits

8.1 Feasibility

- **Technical Feasibility**: Built using proven open-source AI/ML, PostgreSQL, and geospatial tools, functioning as a non-disruptive intelligence layer over existing e-SAKSHI data.
- **Economic Feasibility**: A scalable software solution that limits manual inspection workload and infrastructure costs.
- **Operational & Governance Feasibility**: Uses role-based escalation and provides clear, evidence-backed dashboards to strengthen accountability.

8.2 Impact & Benefits

- Transitions administration from reactive checking to proactive risk identification.
- Reduces manual inspection overhead by prioritizing high-risk targets.
- Identifies potential cost deviations and duplicated works early, aiding in better fund monitoring.

- Enables evidence-based decision making for district authorities and MPs.

9. Limitations and Constraints

To maintain credibility, the system acknowledges several real-world constraints:

- **Data Quality:** Missing or inconsistent e-SAKSHI data can skew results. The system handles this via a "Data Confidence" metric rather than inventing data.
- **Contextual Variations:** Project costs vary by terrain and location. Universal thresholds may cause false positives, so category-aware comparisons are utilized.
- **Model Validation:** The prototype currently runs on representative/mock data. Production deployment requires validation against authorized historical government datasets to prevent model drift and ensure true accuracy.
- **No Confirmed Fraud Classification:** The AI flags anomalies, but human verification is strictly required as anomalies may have legitimate real-world explanations.

10. Future Scope

The platform can be expanded into a government-grade platform through:

- **Authoritative Integration:** Secure API integration with live e-SAKSHI/MPLADS data.
- **Advanced Document AI:** Deeper OCR parsing for invoices, bills, and completion documents to automate arithmetic and consistency checks.
- **Project Health Passports:** Tracking the evolution of a project's risk timeline throughout its entire lifecycle.
- **Smart Inspection Prioritization:** Recommending the specific type of inspection required (e.g., document review vs. field verification) based on the exact nature of the AI alert.
- **Continuous Learning:** Utilizing the human reviewer's "Resolved" or "Escalated" feedback to automatically recalibrate and improve ML detection thresholds over time.

11. Conclusion

MPLADS Sentinel offers an innovative, AI-assisted approach to public infrastructure monitoring. By aggregating multiple risk signals—ranging from NLP-based duplicate detection to financial anomaly tracking—it scores projects and provides easily understandable, explainable alerts to government officials.

Its core USP is clear: **It does not replace human verification; it helps authorities know exactly *where* to look *first*.** By bridging the gap between raw operational data and actionable risk intelligence, Civic Sentinel ensures that public funds are utilized efficiently and transparently.